

PRODUCT REQUIREMENTS DOCUMENT

MVP: AI Egress Compliance Checker

Internal Tool - UAE Fire and Life Safety Code, Travel-Distance & Exit Compliance, v0.1

Version 1.0 - MVP scope for build kickoff
August 2026

1. Summary

Field	Value
Product name (working)	Egress Checker MVP
Owner	You (product/domain) + technical lead
Primary user	Your consultancy's own FLS reviewers (internal use only)
One-line goal	Upload a floor plan, get every travel-distance and exit-count violation flagged against the UAE Fire and Life Safety Code, in minutes instead of hours
Timeline target	8-12 weeks to first internal use on a real project
Explicitly NOT in this MVP	Occupant load, fire-rating, sprinkler/detector checks, external customers, Saudi code, multi-tenant billing

2. Why This Scope (and not more)

This MVP deliberately covers ONE code (UAE Fire and Life Safety Code), ONE building type (single-tenant commercial office floor plate, to limit occupancy-classification edge cases), and TWO checks: maximum travel distance to an exit, and minimum number of exits per floor based on occupant capacity. Everything else from the full platform PRD is deferred. The reason: this is the smallest slice that (a) is genuinely useful to a working consultant today, (b) forces you to solve the hard extraction problem once, and (c) produces real labeled data to justify every later phase.

3. User & Core User Story

3.1 Persona

Internal FLS Consultant - an engineer at your consultancy doing first-pass review of a client's architectural floor plan before deep manual analysis.

3.2 Core user story

- As a consultant, I upload a floor plan (PDF or DXF) for a commercial office floor, so that I can see every point on the plan that exceeds maximum travel distance to an exit, and every zone that doesn't have enough exits for its occupant load, before I start my own manual review.

3.3 Secondary user stories

- As a consultant, I want to see flags overlaid directly on the drawing (not just a text list), so I can immediately locate the problem area.
- As a consultant, I want each flag to show the exact code clause it violates and the measured value vs. the limit, so I can trust and defend the finding to a client.
- As a consultant, I want to mark each flag as confirmed, false positive, or fixed, so my feedback improves the tool and creates a record for the project file.
- As a consultant, I want to export a simple summary (PDF or spreadsheet) of all flags for a floor plan, so I can attach it to my working file.

4. Scope

4.1 In scope

- File upload: single-floor PDF or DXF drawing of a commercial office layout.
- Extraction: walls, doors, corridors, stairs/exits, and room boundaries from the uploaded drawing.
- Rule check 1 - Travel distance: compute the walking distance from every occupied point/room to the nearest compliant exit; flag any path exceeding the UAE Fire and Life Safety Code's maximum allowed travel distance for the occupancy type.
- Rule check 2 - Exit count/capacity: estimate occupant load per floor from area and occupancy-type defaults, and flag if the number/width of exits is insufficient for that load per code.
- Visualization: flags overlaid on the original drawing image, plus a side list with clause citation and measured value vs. limit.
- Reviewer feedback loop: accept/reject/edit per flag, stored against the project.
- Export: downloadable PDF/CSV summary of all flags for a floor.
- Basic project management: create a project, upload one or more floor drawings to it, see review status per floor.

4.2 Explicitly out of scope for MVP

- Occupant-load-driven checks beyond exit count (e.g., corridor width, door swing direction, dead-end corridor length) - Phase 2.
- Fire-rating, compartmentation, sprinkler/detector coverage checks - Phase 2.
- Saudi Building Code or any code beyond UAE Fire and Life Safety Code - Phase 2.
- Residential, healthcare, industrial, or other occupancy types beyond commercial office - Phase 2 (these have materially different rule sets).
- External customers, multi-tenant isolation, billing, self-serve signup - Phase 3/4.
- IFC/BIM ingestion (DXF and PDF only for MVP) - Phase 2.
- LLM-generated plain-language recommendations - MVP shows the deterministic flag and clause citation only; natural-language "suggested fix" phrasing is Phase 2, once there's enough reviewed output to ground it safely.

5. Functional Requirements (detailed)

5.1 Ingestion

ID	Requirement	Priority
ING-1	User can upload a PDF or DXF file up to 100MB via the web app	Must
ING-2	System detects drawing scale automatically from title block/scale bar where present; otherwise prompts user to confirm scale manually	Must
ING-3	System validates the file is a single floor plan (not a full multi-sheet set) and rejects/warns if not, for MVP simplicity	Must
ING-4	User can tag the floor's occupancy type from a fixed dropdown (MVP: "Commercial Office" only, built extensibly for future types)	Must

5.2 Extraction

ID	Requirement	Priority
EXT-1	For DXF input, parse vector layers directly to identify walls, doors, and room boundaries	Must
EXT-2	For PDF input, run computer-vision detection (fine-tuned model) to identify the same elements from the rasterized page	Must
EXT-3	System identifies exit doors/stairs specifically (vs. regular interior doors) using labels and geometry heuristics, with a manual override if misclassified	Must
EXT-4	System builds a walkable path graph of the floor (rooms and corridors as nodes, connections as edges) for distance calculation	Must
EXT-5	User can manually correct any misdetected element (add/remove/move a wall, door, or exit) before running the compliance check	Should

5.3 Compliance checks

ID	Requirement	Priority
CHK-1	Calculate shortest walkable path distance from every room/zone centroid to the nearest identified exit	Must
CHK-2	Compare each calculated distance to the applicable maximum travel distance clause for the tagged occupancy type; flag exceedances	Must
CHK-3	Estimate occupant load per floor/zone using code-defined area-per-person factors for the tagged occupancy type	Must
CHK-4	Compare estimated occupant load to the number and total width of identified exits; flag if insufficient per code	Must
CHK-5	Every flag stores: the code clause reference, the measured value, the code limit, and severity	Must

5.4 Review UI

ID	Requirement	Priority
UI-1	Drawing viewer shows the uploaded floor plan with detected elements and flags overlaid as colored markers	Must
UI-2	Clicking a flag on the overlay highlights the corresponding entry in a side-panel list, and vice versa	Must
UI-3	Side-panel flag entry shows clause reference, measured value vs. limit, and severity	Must
UI-4	Reviewer can set each flag's status: Confirmed / False Positive / Resolved, with an optional note	Must
UI-5	Project dashboard lists all floors uploaded, their review status, and flag counts	Should
UI-6	Export button generates a PDF and/or CSV summary of all flags and their reviewer status for a floor	Must

6. Data Model (MVP)

Entity	Key fields
Project	id, name, client_name, created_by, created_at
Drawing	id, project_id, file_url, file_type (pdf/dxf), occupancy_type, scale, upload_status, uploaded_at
ExtractedElement	id, drawing_id, type (wall/door/exit/room), geometry (coordinates), confidence_score, manually_edited (bool)
Room	id, drawing_id, boundary_geometry, occupancy_type, estimated_occupant_load
ExitPath	id, drawing_id, room_id, nearest_exit_id, calculated_distance, path_geometry
Violation	id, drawing_id, type (travel_distance/exit_count), related_room_id, code_clause_ref, measured_value, limit_value, severity, status (open/confirmed/false_positive/resolved), reviewer_note
Review	id, drawing_id, reviewer_id, status, completed_at

7. Non-Functional Requirements (MVP-scoped)

- Accuracy target: manual spot-check of at least 90% element-detection accuracy (walls/doors/exits) on the DXF path before relying on it for real project review; PDF/CV path accuracy tracked separately and clearly flagged as lower-confidence in the UI.
- Performance: a single floor plan should process (upload to flags-ready) in under 5 minutes.
- Every flag must be traceable to a specific code clause reference stored in the system, not a paraphrase generated on the fly - MVP uses a small hand-curated table of the relevant UAE Fire and Life Safety Code clauses, not a full RAG pipeline yet.
- Data handling: drawings are client-confidential; store in encrypted object storage even for this internal-only MVP, since the habit and the pipeline should carry forward unchanged into Phase 3.
- No AI output is client-facing without a named reviewer's sign-off recorded in the Review entity.

8. Acceptance Criteria / Definition of Done

- A consultant can upload a real (anonymized or live, with permission) commercial office floor plan in DXF or PDF format.
- The system returns travel-distance and exit-count flags, each with a clause citation and measured-vs-limit values, overlaid on the drawing.
- A reviewer can mark each flag's status and export a summary document.
- On a test set of at least 10 real or representative floor plans, the tool's flags match a senior consultant's manual review with no missed life-safety-critical violations (false negatives are the critical failure mode to catch before real use; false positives are acceptable and expected at MVP stage).
- Total time from upload to reviewed, exported summary is measurably faster than the fully manual baseline for the same drawing.

9. Build Plan - 8 to 12 Week Sprints

Sprint	Focus	Exit criteria
1-2	Project setup, file upload, DXF parsing (ezdxf) for walls/doors/rooms	Can upload a DXF and see extracted walls/doors/rooms rendered back in the UI
3-4	Exit/stair identification + walkable path graph (NetworkX) + travel-distance calculation	Given a parsed floor, system computes and displays distance-to-nearest-exit per room
5	Hand-curate the relevant UAE Fire and Life Safety Code clause table (travel distance + exit count/capacity) and wire it into a simple rules check	Travel-distance and exit-count violations are correctly flagged against real clause limits on test drawings
6	Review UI: overlay flags on drawing, side-panel list, accept/reject/note workflow	A reviewer can walk through all flags for a floor and record decisions
7	PDF/CV fallback path for scanned/flattened PDF drawings (bootstrap model, lower-confidence flagging)	A PDF-only floor plan (no DXF available) produces usable, clearly-confidence-flagged results
8	Export (PDF/CSV summary), project dashboard, manual element-correction tool	Full upload-to-export flow works end to end on a real project
9-10 (buffer)	Run the 10-floor-plan accuracy benchmark against senior consultant review; fix false negatives found	Meets the Section 8 acceptance criteria before first live project use

10. MVP Technology Stack (narrowed from the full platform stack)

Layer	MVP choice	Note
Frontend	Next.js + TypeScript, Tailwind, PDF.js + Konva for drawing overlay	Same as full platform - no reason to change later
Backend	Python FastAPI	ML-heavy pipeline; keep it in one service for MVP instead of full microservices
CAD parsing	ezdxf	Primary path for MVP - prioritize DXF input over PDF where possible for accuracy
PDF/CV fallback	PyMuPDF + OpenCV + a fine-tuned YOLOv8 model (start from an open floor-plan dataset, refine with your own labeled drawings)	Treat as secondary/lower-confidence path for MVP
Graph/geometry	NetworkX (path-finding) + Shapely (areas/geometry)	Deterministic - this is the core of the MVP's trust
Database	PostgreSQL + PostGIS	Same schema direction as the full platform PRD, just fewer tables populated
Storage	S3-compatible bucket (in-region if available)	Encrypt at rest from day one
Rules content	A hand-curated JSON/YAML table of ~15-25 relevant code clauses (travel distance + exit count by occupancy type), not a full RAG system yet	Full RAG/LLM explanation layer is explicitly deferred to Phase 2 per the platform PRD
Hosting	Single-region cloud (AWS or Azure, UAE/Bahrain if available); no multi-tenant infrastructure needed yet	Keep infrastructure minimal - this is an internal tool for now

11. Risks Specific to This MVP

- Exit/stair misclassification is the highest-impact failure mode - a missed exit silently breaks every downstream distance calculation. Mitigate with the manual-correction tool (EXT-5) and mandatory reviewer sign-off before any output is trusted.
- Scale-detection errors on DXF/PDF files will silently produce wrong distances. Mitigate by always surfacing the detected scale to the reviewer for a one-click confirm before running checks.
- Hand-curated clause table (Section 10) will need a real update process even during MVP if the code changes - keep it in a version-controlled, easily editable format from day one so this isn't a rebuild later.

12. What Happens After MVP

Once Section 8's acceptance criteria are met and the tool is genuinely saving your consultants time on real projects, proceed to Phase 2 of the platform PRD: add occupant-load-driven checks beyond exit count, fire-rating/compartimentation checks, the LLM/RAG explanation layer, and Saudi Building Code coverage - in that order, using the same staged-scope discipline that shaped this MVP.